

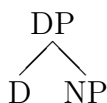
Chapter 7

Functional categories II — the DP

So far in this book we have paid most attention to the structure of the clause, and the way that the requirements of lexical items are met via Merge and Move. In this chapter, we will focus instead on the structure of the phrase which has, as its lexical core, a nominal element, rather than a verb. We will argue that, just as the verb in a clause is surmounted by a series of functional projections, so is the noun. These functional projections in both cases add non- θ -related semantics and have the syntactic effect of providing positions for movement of both heads and phrases. The theory we will develop, then, implies that the true head of a nominal phrase is a functional category, which is the topmost of a series of functional categories, just as T is the topmost of a series of functional categories in the clause.

More specifically, we will see that the topmost functional category in the nominal phrase is that which is usually, in English, filled by a determiner. This means that we will be arguing for the idea that noun phrases are really **determiner phrases** and that they have the following kind of structure:

(1)



Here DP stands for a phrase headed by an element of category D.

We shall see that, just as TPs contain other functional elements than just T, DPs also have a richer functional structure. We will also see that this functional structure is responsible for checking case features, and for providing landing sites for phrasal and head movement processes.

7.1 The category D

In this first section we will concentrate on the syntactic behaviour of determiners, as a preliminary to our argument that these constitute the head of nominal phrases.

7.1.1 Types of D

Take a simple phrase like:

- (2) The letter

The word *the* has a set of **distributional equivalents**, that is, words which can slot into the same position in the phrase. The following example shows just some of these:

- (3) The/a/this/that/some/every/each letter

All of these words are required to come before the noun in English:

- (4) *Letter the/a/this/that/some/every/each

If we pluralise the noun, then we find the following as some of the possibilities:

- (5) The/these/those/some letters

The words which are distributional equivalents with *the* are traditionally called **determiners**, and are assumed to have a new category feature — a **D-feature**. In addition to these examples, we also find plural nominals which appear without an overt determiner (**bare plurals**).

- (6) letters (are on the table)

(6) shows that a plural noun in English can stand on its own. Notice that this isn't true with a singular noun:

- (7) *letter (is on the table)

The contrast between the last two examples suggests that the well formedness of bare plural is due to the existence of a **null determiner**. We can capture the contrast by saying that English possesses a null plural determiner, there is no null singular determiner. (7) is then ruled out if determiners are obligatory in nominal phrases, a proposal we will return to in section 7.2.6.

The different Ds are traditionally split into subcategories. The determiner *the* is traditionally called the **definite article**, while the determiner *a* is traditionally called the **indefinite article**.

The definite article can be used with both singular and plural nouns, as we saw above. The indefinite article *a*, can only be used with singular nouns:

- (8) a. the letter/ the letters
b. a letter/ *a letters

What we see here is an apparent selectional relation between the D and the number feature of the noun.

The plural equivalent of *a* seems to be *some*, when it is used with a plural noun:

- (9) some letters

However, here things get complicated, since *some* can also be used with a singular (10), and bare plural noun is also used like a plural form of the indefinite (11).

(10) We need some technician to help us.

(11) Show me letters!

As well as the articles *the* and *a*, we also have the **demonstratives**, *this* and *that*, and their plural forms *these* and *those*. These words serve to pick out things in the world via their proximity to the speaker, giving the **proximal** demonstrative *this/these* and the **distal** one *that/those*:

(12) this chair; that chair

(13) these chairs; those chairs

The feature that distinguishes proximal from distal determiners is related to person features. Recall that person features are interpreted as picking out the speaker or the addressee. Proximal demonstratives similarly pick out an element which is considered to be near the speaker or not. Just as we have third person pronouns, which are pronouns used for neither the speaker nor the addressee, some languages have demonstratives for items which are neither close to the speaker, nor close to the addressee. For example, Scottish Gaelic distinguishes the following three demonstratives:

(14) an duine seo
the man this
'This man'

(15) an duine sin
the man that
'That man'

(16) an duine siud
the man yonder
'That man over there'

Unlike the articles, demonstratives can occur without a following noun, in which case they are traditionally called **demonstrative pronouns**:

(17) I ate **that**.

Finally, we have words like *all*, *each* and *every*, which are traditionally called **quantifiers** (we discussed some aspects of the syntax of DPs containing quantifiers in 6). The latter two only occur with singular nouns:

(18) every chair; *every chairs

(19) each chair; *each chairs

The semantic function of **quantification** is to pick out the whole set of chairs, either as a lump sum (*every*), or as a set of distinct individuals (*each*). Other quantifiers include words

like *all*, *both*, *most* and *many*.

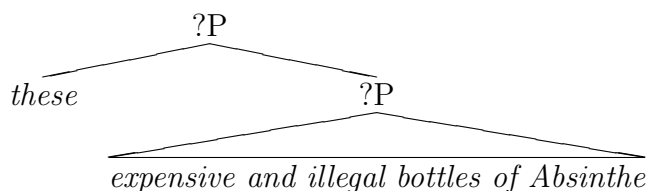
Before we look at determiners from a more theoretical perspective, it's worth pointing out that determiners combine with a (potentially complex) phrasal constituent. We can see this using simple replacement tests:

(20) These expensive and illegal bottles of Absinthe are to be smuggled from Hungary.

(21) These ones are to be smuggled from Hungary.

Here the single word *ones* replaces the string *expensive and illegal bottles of Absinthe* suggesting that it is a constituent, and giving the following schematic structure (I have marked the maximal projections here with question marks, since we will return to what their exact category is later):

(22)



It is also worth pointing out that, although determiners come before the noun in English, this is not true for all languages. For example, in Madi (a Sudanic language) determiners follow the noun:

(23) ebi re/di
 fish the/this
 'The/this fish.'

This latter fact is reminiscent of the variation we saw between head-initial and head-final structures in the VP (that is, between VO and OV languages).

Summarizing, we have seen that there is a class of words which appears prenominal, and which can be subclassified into articles, demonstratives and quantifiers. We expressed the commonality between these words by assuming that they had a shared category feature D. In the next section, we will look at the semantic relation that determiners bear to the rest of the phrase that they occur in. We will see that this relation is not one of θ -role assignment, a fact which suggests that determiners, like auxiliaries, are functional categories.

7.1.2 The semantics of Ds

The most important fact for a semantic analysis of these determiners is that they do not have a θ -related semantics. That is, they do not assign θ -roles to arguments, as, say, Ns or Vs do. Rather, their semantics appears to have to do with restricting the range of referents picked out by the nominal that they (may) occur with.

Take, for example, the definite article. Definiteness is a complicated phenomenon, but one useful way to think about it is that it links the nominal phrase to the context of the utterance in a particular way. More specifically, it has to do with whether the phrase in

which the article appears describes, or picks out, something which the speaker can assume the hearer is **familiar** with, or not. In English, if the speaker can assume that the person he or she is talking to has already come across the thing referred to, then a definite determiner is used, otherwise an indefinite is used.

For example, in the first of the following two conversations, the use of the definite article in the italicised phrase seems odd, because the cocktail shaker hasn't already been mentioned, whereas in the second, the definite article seems perfectly appropriate:

- (24) Speaker A: Are you having a party this year?
 Speaker B: Yeah, tonight. I'm just off to buy *the new cocktail* shaker
- (25) Speaker A: Did you buy a new cocktail shaker and some gin for the party?
 Speaker B: Yeah, *the new cocktail shaker* was really expensive, though.

The oddness of (24) here doesn't have to do with grammaticality. Speaker B's sentence is perfectly well formed, and if the indefinite determiner *a* was used in place of *the*, the sentence would be acceptable in this context. The sentence is odd in this conversation because the definite determiner adds something to the sentence which has to do with each speaker's knowledge of the background to the conversation. No 'new cocktail shaker' has been mentioned, and so using the definite determiner with this phrase leads to **infelicity** (pragmatic unacceptability). Similarly, the indefinite determiner signifies that the referent of its phrase is to be considered something new in the discourse. This is why, if we swap in *a* for *the* in (25), the discourse again becomes disjointed: the use of *a* is infelicitous here, because there is indeed a 'new cocktail shaker' which is familiar to both participants in the conversation.

Notice that you can sometimes get away with using the definite article even if the particular thing hasn't yet been mentioned, but its existence, and therefore its familiarity to the hearer, can be inferred from what's been said. This phenomenon is known as **accommodation**:

- (26) Speaker A: Did you buy stuff for the party?
 Speaker B: Yeah, *the cocktail shaker* was really expensive, though.

Definiteness, then, is something which helps to integrate an utterance into a conversation. It does this by giving clues to the hearer about what the speaker thinks he/she already knows. Linguistic devices that do this kind of work are said to have a **pragmatic effect**.

Demonstratives have a similar semantics to articles, in that they signify how the speaker wants to direct the hearer's attention: demonstratives signify that the speaker assumes that the hearer will be able to identify the referent of the phrase as being close or far away, in either spatial or in discourse terms.

Quantification, however, serves a different semantic purpose. Whereas definiteness signifies a discourse property of the referent of its phrase, quantification doesn't pick out a referent at all. To see this, take an example like (27).

- (27) No vampire can survive sunrise.

Clearly the phrase *no vampire* here has no referent. You can't point to the thing that the

phrase *no vampire* picks out, because it doesn't pick out anything. The same is true for phrases like *each vampire* or *every vampire*. These phrases don't have a referent in the usual sense of the word.

What, then, is the semantics of these quantifiers? The easiest way to think about this is to look at quantifiers in whole sentences:

(28) No vampire slept.

(29) Every vampire slept.

In (28), what we are saying is that the set of entities picked out by the predicate *vampire* and the set picked out by the predicate *died* are disjoint: they do not have even a single element in common. There may be lots of vampires around, and there may be lots of sleeping things, but these sets do not overlap. The same kind of analysis can be developed for *every* in (29). In this case, the set of vampires is asserted to be completely included in the set of sleeping things.

It is interesting, then, that we have two completely different semantic functions (familiarity and quantification) carried out by elements with the same syntactic category (D). However, what is crucial for our purposes here is that the semantics of elements of category D is not θ -related.

We have come across categories which bear a non- θ -related semantics to other categories already: this is the relation between T and *vP*, for example. By analogy, it appears that D, like T, is a functional category.

In the next section we will focus on how to syntactically represent Ds, so as to capture this property.

7.2 Ds as Heads

7.2.1 Complementary Distribution of Ds

The first important syntactic thing to notice about this set of words is that they are in **complementary distribution**. What this means is that, because they have exactly the same distribution, they compete for the same position in the structure of the phrase.

(30) *The this man

(31) *Each a man

(32) *Some those letters

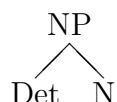
(33) *The the man

Early approaches to generative grammar assumed that phrases were structured by having certain slots in them. This was done by using phrase structure rules like those we saw briefly in chapter 1. A noun phrase would have a single determiner slot, and a single noun slot, and its structure would be specified by a rule like the following:

(34) $NP \rightarrow \text{Det } N$

This rule would generate trees like the following:

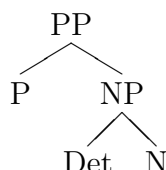
(35)



Likewise, in a prepositional phrase there would be one slot for a preposition and one for the noun phrase.

(36) $PP \rightarrow P\ NP$

(37)



Since a single node in a tree can accommodate only one element, it follows that a noun phrase could have only one determiner and only a single noun. Similarly, it would be impossible to have two or more prepositions heading a single PP, or more than one NP complement of the preposition. All of these properties are stipulated in the phrase structure rules. In this way, the notion of complementary distribution, and the particular case of complementary distribution of Ds, is automatically dealt with.

However, the approach to syntax we have been developing here does not make use of these structure specific rules. Instead, it appeals to general principles, such as the universal Hierarchy of Projections, and to the operations Merge and Move. We would like to develop an analysis of the syntax of Ds in these terms.

Putting our discussion from the last few sections together, the general properties of elements which are of category D and which we want to capture with a syntactic analysis are the following:

1. There seems to be a selectional relationship between D and N in terms of number.
2. Ds may occur on either side of their related Ns in different languages, in a way which is reminiscent of linearization of heads and their complements.
3. Ds combine with a complex constituent.
4. Their semantics is related to notions of familiarity, quantification and proximity, rather than to notions of θ -role assignment.
5. Ds, in English, occur in complementary distribution.

We will tackle the last of these factors first.

7.2.2 Capturing the Complementary Distribution

The approach we have been outlining so far deals with complementary distribution of functional categories via the Hierarchy of Projections. The Hierarchy states the category and position of various functional categories in the clause: the reason that modals and non-finite *to*, for example, are in complementary distribution is that they both have the category feature T, and the Hierarchy of Projections only provides one ‘slot’ for elements of this category in a clause.

We can transfer this idea from the clausal to the nominal domain. We have already motivated the idea that Ds are functional categories, just like T, Neg, Perf, Prog and Pass: they do not have a θ -related semantics. Just as functional categories in the clausal domain are determined by the Hierarchy of Projections, so we would expect functional categories in nominal expressions to be determined by a nominal version of the hierarchy.

We will explore this idea and state a new Hierarchy of Projections, which appears above nominal rather than verbal lexical categories:

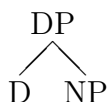
(38) **Hierarchy of Projections**

- a. *Clausal*: T $\rangle$ (Neg) $\rangle$ (Perf) $\rangle$ (Prog) $\rangle$ (Pass) $\rangle$ *v* $\rangle$ V
- b. *Nominal*: D $\rangle$ N

Recall the way we developed the clausal Hierarchy of Projections from a simple to a more elaborate structure. As you might have guessed, the same fate will befall the nominal Hierarchy.

If this idea is on the right track, then we have the following structure for nominal expressions:

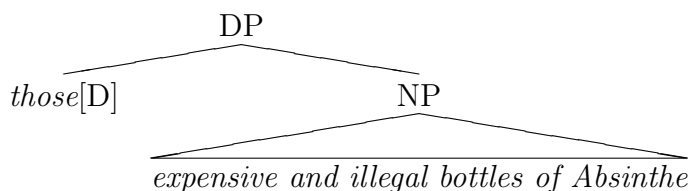
(39)



This proposal immediately captures the complementary distribution of Ds. Since there is only one D slot, there can be only a single determiner. By its very nature, it also captures the non- θ -related semantics of Ds. Elements of the Hierarchy of Projections do not bear a θ -theoretic selectional relationship to each other.

The new approach to nominals also correctly accounts for the fact, discussed above, that Ds combine with complex constituents. The NP node in (39) is phrasal by the general principles of our system, since it projects no further. Phrasal categories are capable of being complex and it is therefore no surprise that we find a syntactically complex constituent here. The structure of our previous example is then:

(40)



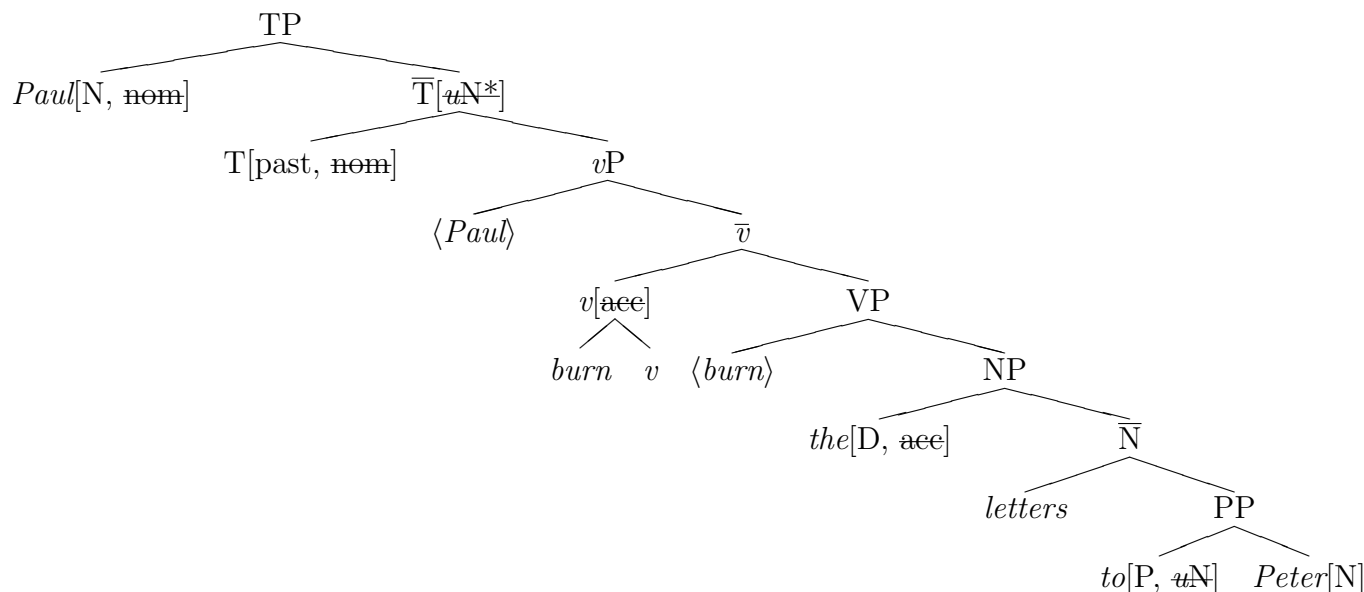
Furthermore, as we have already seen, the linearisation of heads and their complements may vary across languages. Since NP is complement to D, under our new analysis, it is expected that the linear order of D and N should vary, with some languages having DN order, and others ND (such as the Madi example discussed above).

7.2.3 Implications of this Analysis

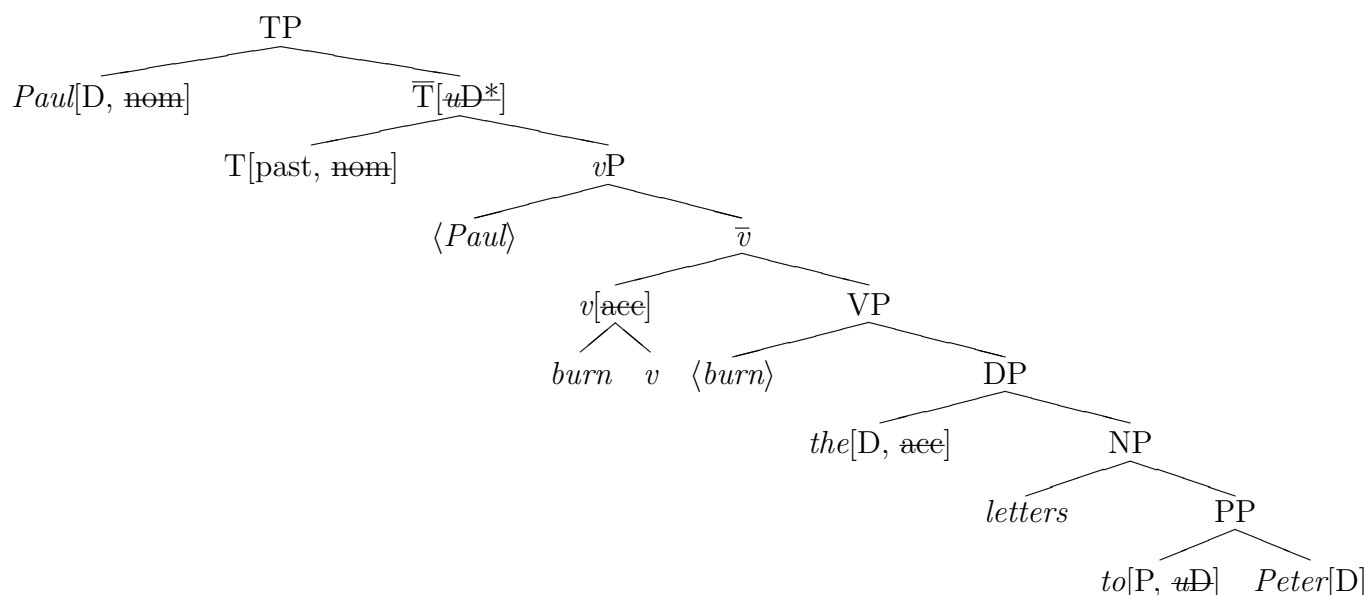
However, this approach threatens an idea that we have maintained since Chapter 3: the idea that N is the head of nominal phrases. It will also mean that all of the cases we have been treating as NPs, will have to be recategorized as DPs, and that selectional N-features (including the EPP feature of T) will have to be recategorized as selectional D-features (so that the EPP feature is $[uD^*]$ rather than $[uN^*]$).

Similarly, verbs and prepositions, which we said had selectional N-features, must rather have selectional D-features. A typical tree will then be represented as (42), rather than as (41).

(41)



(42)



This recategorisation does not mean that we lose anything real in the system. The distributional motivation for why phrases of the same type occur in the same position can be easily stated in terms of D-features rather than N-features. Notice that NPs still exist in our new system, but they are now complements of D; this is as expected since we have morphological motivation for the existence of an N-feature, and Ns clearly combine with elements such as PPs to make larger phrases where N projects.

We conclude then that our previous analysis of nominal phrases as NPs is too simplistic. NPs do exist, but they are the complements of a functional category. Ds are the heads of nominal phrases.

At this point in the chapter, it would be a good idea to redraw all the trees you constructed in doing the exercises for the last chapter. You should use our new analysis of nominals as DPs, and note carefully how that affects the selectional features and EPP features in the trees.

7.2.4 Further Questions

Bare Nominal Phrases

With this much in place, we can now turn to some further questions raised by the DP analysis. The first that we will consider is how we deal with bare plural nominals like *letters*. If these are simply nouns, then they will not satisfy the selectional requirements of verbs (since these are now stated in terms of c-selectional D-features, rather than c-selectional N-features). This will predict the ungrammaticality of such obviously grammatical examples as (43):

- (43) a. I wrote letters.
b. We ate jellyfish.

The verb *write* c-selects a DP, under our new approach. We need to say this since we can have the following:

(44) I wrote the letters.

But if *write* c-selects a DP in (44), then it should be c-selecting a DP in (43). It follows that the bare nouns *letters* and *jellyfish* are DPs, even though they contain no overt D.

Note that in many other languages corresponding examples with bare nouns are indeed ungrammatical, as the following examples from French show.

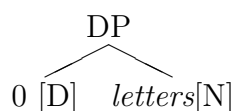
(45) *J'ai écrit lettres
I-have written letters
'I have written letters'

(46) J'ai écrit des lettres
I-have written some letters
'I have written letters'

In languages that behave like this, we simply say that no null D is available.

Putting this together with our previous argument, it appears that English bare plurals are really DPs, and that they have a null determiner, as required by the Hierarchy. This is the conclusion we also reached in 7.1.1:

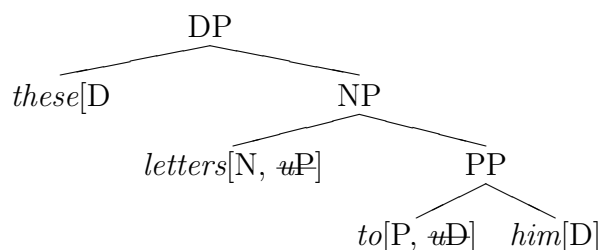
(47)



Pronouns

We can incorporate pronouns into our new system by treating them simply as Ds. When a pronoun is merged into the tree, it receives a θ -role and is selected, and so automatically it has phrasal status:

(48)



This approach also makes sense of examples like the following:

(49) [We linguists] love to argue

(50) [You friends of the king] are all the same

Here we see the pronoun, which is a D, exceptionally taking an NP complement.

7.2.5 Possessors — Specifiers of DP

We have, so far, developed an analysis for nominal phrases which treats them as DPs, rather than as NPs. Since, on this analysis, D is a head, we might expect it to have a specifier, as well as a complement. In this section we will propose that certain possessor phrases in English and other languages appear as the specifier of DP.

Possessor semantics, in English DPs, is syntactically realised in two main ways: via a preposition (51), and via a construction traditionally known as the **Saxon Genitive** (52):

(51) An idea of Evan's.

(52) Evan's idea.

We will concentrate here on the latter construction.

The presence of a Saxon genitive is blocked by the appearance of an overt article in English:

(53) *The Evan's idea

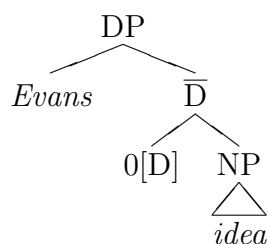
(54) *Evan's the idea

One interpretation of this might be that articles and Saxon genitives are in complementary distribution, and hence compete for the same position in the DP. However, this would be inconsistent with what we have said already: our approach treats Ds as X^0 elements, but Saxon genitives can clearly be XPs:

(55) [One of our oldest friend's] idea

Another approach to this problem, consistent with what we have said already, is to treat the Saxon genitive as the specifier of a null D:

(56)



We have already met one null D which is used with plural Ns; here we would say that there is a second null D which occurs with a Saxon genitive specifier.

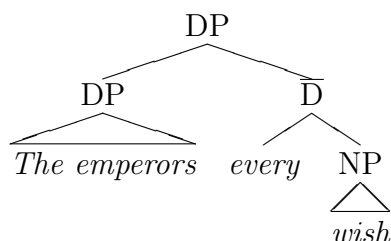
Is there any evidence that this is the right approach? In fact, there is actually an overt determiner in English which is not in complementary distribution with the Saxon genitive: *every*.

(57) Evan's every idea was completely insane.

(58) The Emperor's every wish was immediately carried out.

These examples show that there needs to be enough structural space within the DP to host an overt determiner and a Saxon genitive. the structure is as follows:

(59)



his analysis of *every* naturally extends to the other cases of saxon genitive's in English if we adopt the idea that the latter contain a null D.

The analysis outlines here receives some crosslinguistic backup from the following examples from Hungarian:

- (60) Peter minden kalap-ja
 Peter's every hat-[DEF]
 'All Peter's hats'
- (61) Peter ezen/azon kalap-ja
 Peter's this/that hat-[DEF]
 'This/that hat of Peter's'
- (62) Peter valamannyi kalap-ja
 Peter's each hat-[DEF]
 'Each hat of Peter's'

An approach which places prenominal possessors in the specifier of DP makes immediate sense of such data. Whereas all Hungarian Ds allow a specifier, only *every* and a null D do in English.

7.2.6 More Null Ds

We have now seen many examples of overt (that is, pronounced) determiners, and two possible examples of covert (that is, silent, or null) determiners: one is found with plural nouns, and the other with genitive possessors:

- (63) I have sent 0 letters to Environmental Heath.
- (64) The microphone-salesman's 0 irritating patter was relentless.

We have also seen that it is impossible to have a null determiner with singular nouns like *letter*:

- (65) *I have sent 0 letter to Environmental Heath.

(65) is accounted for because *send* c-selects a DP, but there is no null D which combines with a singular noun in English.

This is a purely English-specific fact. There are other languages which allow null Ds to appear with singulars, as the following example from Gaelic shows:

- (66) Chuir mi litir dha
 send-PAST I letter to-him
 ‘I sent a letter to him.’

In this example the bare singular *litir*, ‘letter’ occurs with a null determiner which means that it projects to a DP. This DP can then satisfy the c-selectional requirements of *chuir*, ‘sent’.

Mass and Count

We do, however, find a singular null determiner with a certain class of nouns (italicised here) in English:

- (67) Andrew likes *lard* on his sandwiches
 (68) *Oil* spread over the sea-shore.
 (69) The wizard turned the beetle into *beer* with a wave of his wand

These nouns, *lard*, *oil*, *beer*, are so called **mass-nouns**, and contrast with **count-nouns** like *letter*, *beetle*, *sandwich*. Mass nouns also contrast with count nouns in not pluralising easily:

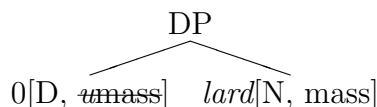
- (70) lards; oils; beers

These words mean types of lard, types of oil and types of beer respectively. It seems impossible to make the actual noun itself semantically plural, because mass nouns denote substances which are inherently uncountable. This means that syntactic marking for the feature plural is interpreted instead in an odd way. The distinction between mass and count nouns appears to be a semantic feature which the syntax is sensitive to.

This allows us to say that we have a null determiner which agrees with its complement in terms of mass status. We can implement this easily by specifying an uninterpretable feature on the null D which will be checked only if the D c-commands an appropriate nominal:

- (71) D [*u*[mass]]

- (72)



We have now met three types of null D: plural, mass, and the null D that appears with a Genitive specifier.

Proper Nouns

There is an argument for a fourth null determiner in English which occurs with proper names. Notice that, like pronouns, proper names may occur with determiners:

(73) The Peter we all like was at the party

(74) The Paris I used to know is no more

These sentences have slightly odd interpretations: in each case the proper name is assumed to hold of more than one thing in the world, and the determiner is used, together with other modifiers, to clarify which thing is meant.

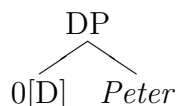
Distributionally, we can see that the subjects of (73) and (74) can be replaced with just the proper name:

(75) Peter was at the party

(76) Paris is no more

What this suggests is that *Peter*, and *the Peter I used to know* are both DPs. One easy way of capturing this is to assume that there is a null determiner in (75) and (76) which takes the proper name as its complement:

(77)



This analysis receives further support from languages where proper names must occur with a determiner, such as the following example from Greek:

(78) O Giorgos ephuge
The George left
“George left.”

We can implement this by specifying that the featural content of the complement NP of the D is [proper]:

(79) D [*u*proper]

Spelling out Null Ds

When we talk about null Ds, it is important to be aware that what we are really saying is the following: the spellout rules for particular feature bundles result in a null phonology. Null Ds, just like everything else in our system, are just spellouts of feature bundles, it’s just that the spellout lacks any phonological value.

We can illustrate this by examining those null Ds that occur with only a plural complement:

(80) a. *man arrived.

- b. Men arrived.

How do we ensure that the (a) example is not generated.

Firstly, we know that some D must be generated. In (80), both examples involve a subject in the specifier of TP. Since T bears the EPP feature [uD^*], the specifier of T must be a DP, or else the EPP feature would not be checked.

So what we need to do now, is ensure that there is no D with a null pronunciation that occurs with singulars.

This is fairly easy to do, given the Agree mechanism we have already motivated. We can say that the indefinite determiner in English has the following feature specification:

- (81) D[*indef*, *unum*:]

Now, when this D combines with an N(P), the interpretable number feature of that N(P) will check and value the [*unumber*] feature of D.

- (82) a. D[*indef*, ~~*unum*~~:*sing*] man[*sing*]
 b. D[*indef*, ~~*unum*~~:*pl*] men[*pl*]

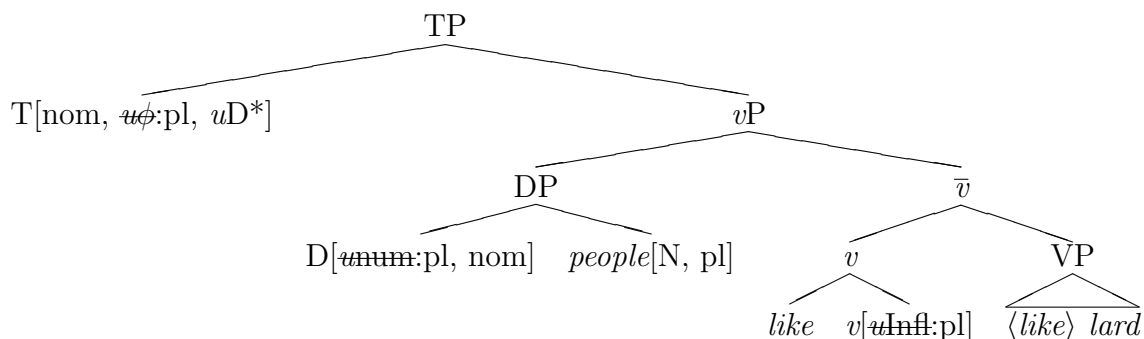
When the spellout rules apply, the D with the valued plural features receives a null pronunciation, while the D with singular features receives a pronunciation as the indefinite determiner, *a*. This system means that it is simply impossible to have an example with a bare singular, because there is no spellout for a singular indefinite determiner as a null element.

Our approach interacts with subject-verb agreement in the right way. To see this, take the following contrast:

- (83) a. People like lard.
 b. *People likes lard.

The null D in both of these examples bears valued [*pl*] features. Although these features have been checked, they can themselves value the ϕ -features on T. Eventually they are pronounced on little *v*, or on the appropriate auxiliary:

- (84)

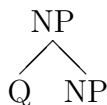


So far, in this chapter, we have established an analysis of nominal phrases which adopts the idea that they are headed by a functional projection categorized as D. This analysis

captures the idea that the semantics of Ds is not θ -related, that Ds Merge with a potentially complex constituent, that there is an agreement relation in number between certain Ds and their NP complement, and the complementary distribution of Ds in English. It also provides us with a plausible structural position for Saxon genitives: the specifier of DP.

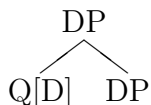
A final advantage is that our analysis of floating quantifiers is now theoretically consistent. The simplified view that we took in the last chapter was that quantifier phrases which could undergo stranding had the following structure:

(85)



We assumed that the quantifiers were heads, and were left with the problem of why they apparently do not project. The DP approach solves this, since these quantifiers do in fact now project. Our revised structure for these floating quantifiers is as follows:

(86)



What is special about them is that they apparently take a DP complement, rather than an NP complement. This allows them to be stranded when their complement raises to the specifier of TP to satisfy EPP. Of course, this now raises a new problem: this structure does not conform strictly to the Hierarchy of Projections, since we have two D nodes. Clearly there is something special about the syntax of these quantifier phrases, but we will not solve the problem here.

In the remainder of the chapter, we turn to what's going on in the lower reaches of DP: how arguments of N are projected, and how their features are checked.

7.3 Θ -role Assignment in DP

In this section we will focus on how arguments of Ns distribute within the DP, trying to tease out similarities with v P, and noting any differences. We will focus firstly on nouns which are transparently related to verbs.

There are many nouns in English which bear semantic relations with DP arguments in the same way as associated verbs do. Take, for example, the verb *analyse*. This is a two place predicate:

(87) The therapist analysed Morticia

The related noun *analysis* intuitively assigns the same θ -roles:

(88) The therapist's analysis of Morticia.

Whereas in the case of the verbal structure we find a nominative and accusative DP (as evidenced if we swap in pronouns to give *She analysed her*), the arguments in the nominal structure have a different syntax.

The Theme argument is realised as a prepositional phrase, while the Agent appears to have the same syntax as a possessor does—the **Saxon genitive**. When the Agent is a pronoun, it appears in its genitive form (see Chapter 2).

We can compare the case properties of clauses and DPs straightforwardly. Whereas in a clause we find nominative and accusative, in a DP this is impossible, and we find genitive, and an *of*-PP:

(89) *He analysis her (was flawed)

(90) His analysis of her (was flawed)

Moreover, it turns out to be the case that there is some unexpected flexibility in the assignment of θ -roles:

(91) The analysis of the problem was flawed

(92) This problem's analysis is made a lot easier when you understand differential equations.

Here we see the Theme argument appearing as a PP (in (91)), but it can also appear as a Saxon genitive ((92)). This kind of flexibility is not observed with Agents:

(93) Morticia's analysis was the most successful

(94) The analysis of Morticia took longer than that of Gomez.

In (94), *Morticia* can only be interpreted as the Theme argument. If there are both an Agent and a Theme in the DP, then the Agent must be a Saxon genitive, and the Theme must be a PP. There are some apparent counterexamples to this generalisation, such as (95):

(95) ?The shooting of the hunters.

In this example, it appears that the DP *the hunters* is being interpreted as the Agent. However, closer inspection reveals that the association between this DP and the noun is much looser. If we put this DP in a sentence, we see the following contrast:

- (96) a. The shooting of the hunters was very loud.
b. The shooting of the hunters took a long time.

The (b) example shows that, if the interpretation of *shooting* is really that of an event, then *the hunters* can only be interpreted as a Theme; this sentence is ungrammatical if *the hunters* is interpreted as an Agent. This suggests that what we are seeing in the (a) example is an interpretation where the hunters are associated in some plausible fashion with the shooting: in a sense, the shooting belongs to the hunters, rather than them being its Agent.

We will assume, then, that our generalisation stands, and we can summarize it as follows:

(97)

	(Saxon) Genitive	PP
Agent	✓	*
Theme	✓	✓

Not all nouns assign θ -roles. For example, simple nouns like *wallet* or *goldfish* have no arguments. As we have already seen, though, they may occur with possessors:

- (98) Morticia's goldfish
- (99) Gomez's wallet
- (100) The goldfish of Morticia's
- (101) The wallet of Gomez's

In none of these examples are the proper names interpreted as Theme or Agent. They are interpreted instead as possessors.

The last two examples are cases where possession is marked with a prepositional phrase. The nominal in this PP is also marked, with something that looks rather like a Saxon genitive. However, a deeper look tells us that this marking is not exactly the same, since pronouns in this position have a different morphological form from pronouns in Saxon genitives:

- (102) My book
- (103) *A goldfish of my
- (104) A goldfish of mine

We will call this special marking that appears in a possessive *of*-PP, the **independent genitive**. The independent genitive is not used to mark Themes, only Possessors:

- (105) *The therapist's analysis of Morticia's

What we have then, are certain nouns which seem to have argument structure and assign θ -roles, while other nouns have no argument structure. For nouns with arguments, these arguments can be realised as a PP headed by *of* when the argument is a Theme, and as Saxon genitive when the argument is an Agent or Theme. Nouns without arguments may have possessors which are realised as an independent genitive or as a Saxon genitive. It is not possible to have two Saxon genitives both applying to the same noun:

- (106) *Morticia's Gomez's wallet

Finally, it also does not seem to be possible to express both an Agent and a possessor in the same DP. Take a noun like *photograph*, which is semantically capable of having an Agent (the person who took the photograph) and a possessor (the person who now owns the photograph). Since the Agent does not have to be the possessor, and since possessors can appear as *of*-PPs with independent genitive marking, it should be possible to have both:

- (107) Morticia's photograph of Pugsly
- (108) That photograph of Pugsly of Morticia's
- (109) *Gomez's photograph of Pugsley of Morticia's

(107) shows that it is possible to have a prenominal Saxon genitive with an Agent interpretation, and (108) shows that it is possible to have a postnominal independent genitive whose interpretation is unambiguously possessive. What does not appear to be possible is to have both together, as in (109). This seems to be a purely syntactic restriction, rather than anything semantic, since it is possible to express the Agent instead as a *by*-PP or the possessor by using an extra modifying adjunct:

(110) That photograph by Gomez of Pugsley of Morticia's

(111) Gomez's photograph of Pugsley belonging to Morticia.

Let us now try to develop an analysis of the syntactic behaviour of arguments within the DP.

7.4 Little *n* and the Low Structure of DP

In this section we will provide the beginnings of an analysis for the fact that Agents do not appear as *of*-PPs in nominals. We will do this by extending our treatment of clausal structure to nominal structure. We will argue that, just as verb phrases are contained in a little *v* projection, noun phrases are contained within a little *n*P. This means that the Nominal Hierarchy of Projections looks as follows:

(112) **Nominal Hierarchy of Projections:** $D \rangle n \rangle N$.

The main theoretical motivation for this kind of approach is that the generalisations we stated above which governed the appearance of arguments within DP are very similar to what would be predicted if we were to generalise the UTAH to nominals. Recall that Agents could not be realised as post-nominal *of*-phrases:

(113) *The imposition of the government of a fine.

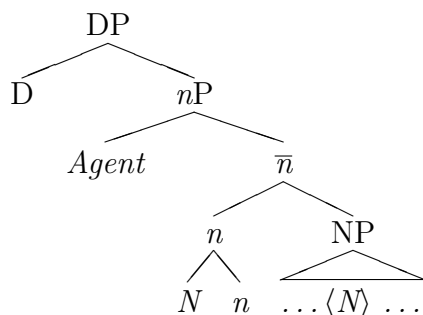
(114) The government's imposition of a fine.

This is parallel to the syntax of verb phrases: the Agent cannot occur as a post-verbal object. This phenomenon follows from the UTAH, which states that arguments Merged as daughters of VP are interpreted as Themes and that to have an Agent interpretation, an argument would have to be Merged as the specifier of *v*.

The UTAH straightforwardly generalizes to the nominal domain. We can say that arguments which are Merged as daughters of NP are interpreted as Themes, and that arguments which are Merged as specifiers of a little *n* projection, are interpreted as Agents.

This would give us a structure for the DP which would look as follows:

(115)



This basic structure gives us a partial explanation for why Agent's do not appear as *of*-PPs. It is for just the same reason that Agents do not appear as objects: they are in the wrong position to have this interpretation. We will see exactly how the UTAH combined with case checking mechanisms gives us the right results in section 7.4.2, but we will leave the explanation at this intuitive level for the moment.

From what we have just said it appears that there is good motivation to extend the UTAH to the nominal domain. However, we have not yet seen any evidence that such a category as little *n* exists. We turn to this in the next section.

7.4.1 Motivation for Little *n*

Just as we argued that Agents in verb phrases are Merged as specifiers of little *v*, we can bring to bear the same kind of evidence, from binding and coordination, that Agents in noun phrases are Merged as specifiers of a little *n*.

It is difficult to come up with clear contexts with which to test reflexive binding, but from the contrasts in the following examples, it appears that the Theme c-commands the Goal in DPs, just as it does in *v*Ps. For the following sentences, the context is one where gladiators are slaves whose masters may free them by giving them to themselves:

(116) The consul's gift of the gladiator to himself.

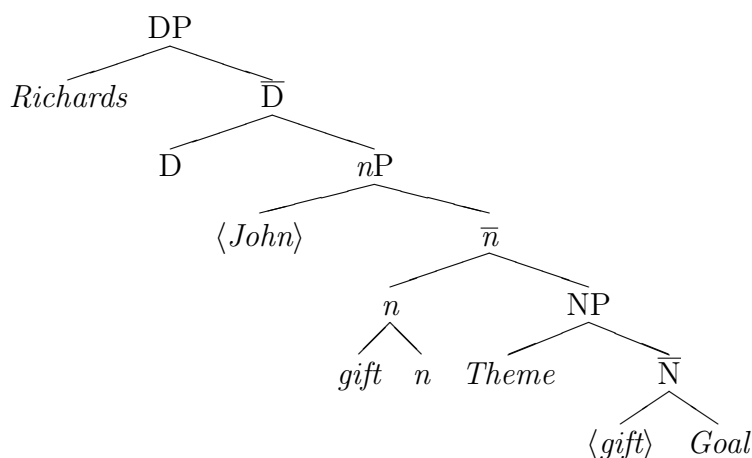
(117) *The consul's gift of himself to the gladiator.

More obvious evidence comes from co-ordination. We find the same pattern within NP as we found within VP:

(118) Richard's gift [of the helicopter to the hospital] and [of the bus to the school].

This suggests that, just as VP is selected by a little *v*, NP is selected by a little *n*, whose function is to introduce an Agent. It is this category that is being coordinated in (118). The structure we adopt is, then:

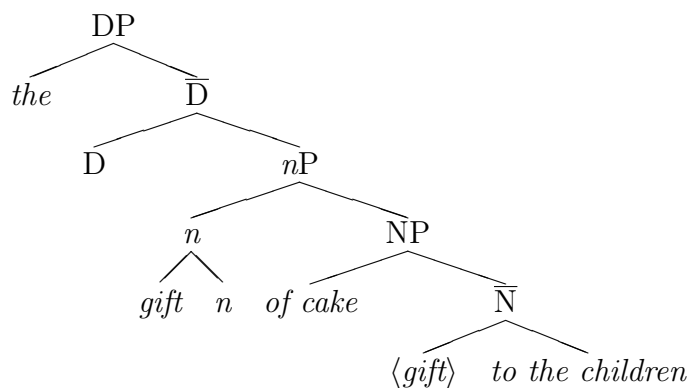
(119)



Once again, in the same way as *V* moves and attaches to *v*, *N* moves and attaches to *n*. I have also represented the idea that *John's* has moved from the specifier of *n* to the specifier of DP, just like the subject moves to the specifier of TP in a sentence. We will return to this movement process in section 7.4.2.

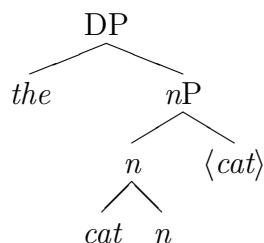
One final question we need to address is whether *n* is projected in all DPs, or just those with Agents. If we look at the noun *gift*, we can see that it need not have an Agent, and we might think about dealing with this by saying that the projection of little *n* is optional. However, note that the noun still precedes both Theme and Goal arguments. If the structure we proposed above is right, then *gift* must raise to *n* to get the right word order, suggesting that *n* is projected even if there is no Agent.

(120)



We will assume, then, that just as little *v* is an obligatory projection, but may lack a specifier, little *n* behaves in the same way. This will mean that even a simple noun like *cat* will have the following structure:

(121)



7.4.2 Case Checking in DP

We have already met the type of feature that regulates the positions of DPs in sentences. Recall that pronouns are specified for different case features depending on their position:

- (122) He loves him
- (123) *Him loves him
- (124) *His loves him
- (125) His book
- (126) *He book
- (127) *Him book

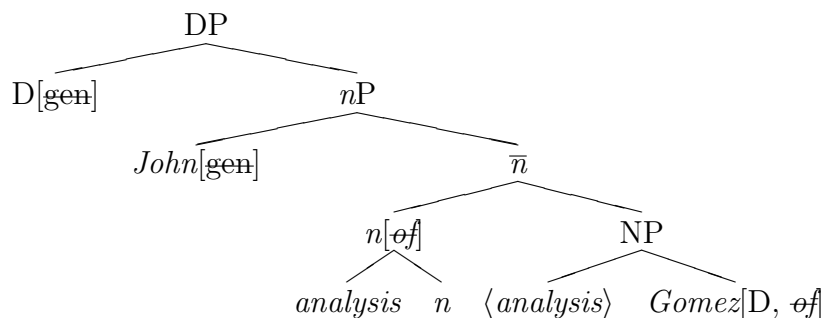
In (123) and (124), we can see that neither an accusative nor a genitive version of the pronoun can go in subject position (the specifier of TP). This slot is reserved for nominatives. Similarly, only a genitive pronoun can occur in the same slot as a possessor phrase in DP, nominatives and accusatives are barred ((126) and (127)). The case feature on a DP is intrinsically linked to the syntactic position of the DP.

In (122), we find nominative case on the pronoun, rather than accusative or genitive because the T node checks case on the DP that ends up in its specifier. We can straightforwardly extend this approach to case checking internal to DP: we find genitive case in DP rather than nominative because the D head in a DP checks genitive case.

Let us now transfer these ideas to our analysis of Saxon genitives in general. The hypothesis we have developed so far is that arguments of a noun are generated in the lexical projection. The Theme is Merged as the daughter of NP and Agents are Merged as the specifier of *n*. We need to account for the fact that Themes may appear as Saxon genitives or *of*-PPs, while Agents only appear as Saxon genitives.

We can do this using the kind of theoretical tools we already developed for sentences. We assume that *n* checks a case feature on the *of*-Phrase which we will just represent as [*of*], while the zero determiner we motivated in section 7.2.5 checks a [gen] feature on the Agent. This gives us the following kind of structure:

- (128)



The [θf] case feature on n is weak, and so the *of*-Phrase stays *in situ*. However, there is evidence that the Agent actually moves to the specifier of DP. Recall that the determiner *every* allowed a Saxon genitive, and that this came before the determiner:

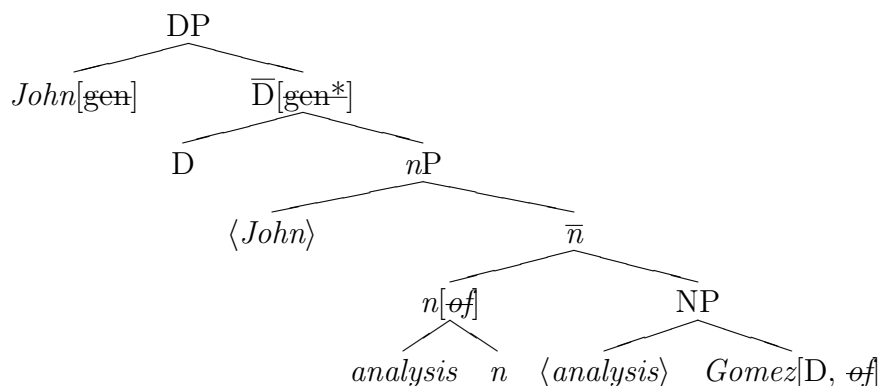
(129) Evan's every desire

This suggests that the Agent DP raises past the determiner into the specifier of DP:

(130) Evan's every $\langle Evan's \rangle$ desire

What is this movement driven by? As usual, we attribute it to the strength of a feature. Now, whereas there was good evidence in the clause that the relevant feature was not case, but rather a special EPP feature, there is no such evidence in the DP. We assume then that the [gen] feature of D (which is optional) is strong, and that movement of the Agent takes place to satisfy the locality requirement imposed by this feature:

(131)



The [gen*] feature on D Agrees with the case feature on John, valuing it. In order to achieve the locality required by the strength of this feature, it projects to the $\bar{D}$ level, and *John* raises to the specifier of DP.

This tree is then spelled out by special morphological and phonological rules. The ones which are of particular interest to us are those that spell out the case features [gen] and [θf]. We can state them roughly as follows:

(132) pronounce [DP, gen] by pronouncing everything within DP and then *s*.

(133) pronounce [DP, θf] by pronouncing *of* and then everything within DP.

Since null D values the case feature of the possessor or Agent, this DP has to appear as a genitive:

(134) The girl's book

(135) *The girl book

(136) Her book

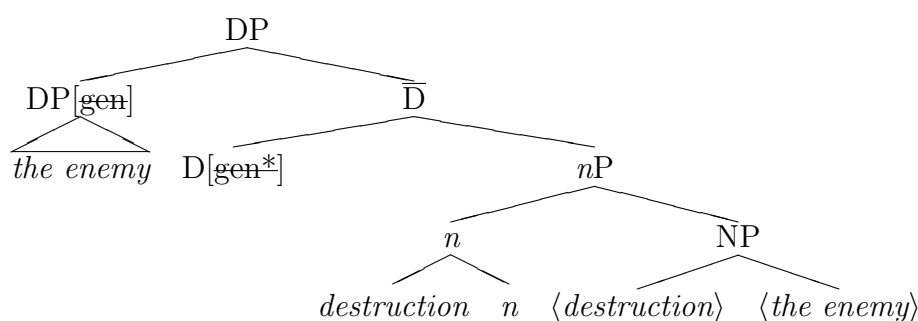
(137) *She book

We can now turn to the fact that Saxon genitives may also be interpreted as Themes. This follows from our earlier claim that little *n* only optionally projects a specifier (that is, its selectional D-feature is optional). If the little *n* is of the sort which projects no specifier, then the [gen*] feature on D will match the case feature of a DP generated as the Theme, correctly predicting that the Theme can raise to the specifier of D and be realised as a genitive:

(138) The enemy's destruction (by the troops)

The derivation for such an example is straightforward and parallels that of an unaccusative. Firstly *the* bearing an unvalued [case] feature is merged with *enemy*. This DP is then merged with *destruction*, thereby obtaining a Theme θ -role. Since there is no overt Agent in the DP, little *n* is Merged but selects no specifier. The zero determiner is then Merged. The strong [gen*] feature on this determiner matches and values the case feature on the Theme, attracting it into the specifier of the whole DP.

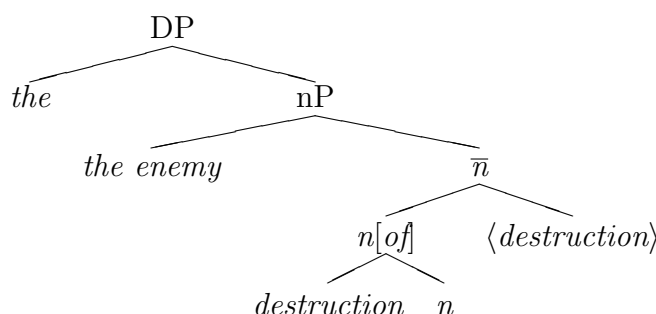
(139)



We have now captured some important facts about the syntax of DPs. Our system correctly predicts that the Agent will never appear as an *of*-Phrase. This is because it is Merged after the little *n* which bears the [of] case feature. Since the *n* head searches in its complement (its c-command domain) for a matching [of] feature, the Agent will never be able to satisfy this requirement. We therefore correctly rule out examples like the following, with an Agent interpretation for the *of*-Phrase:

(140) The destruction of the enemy

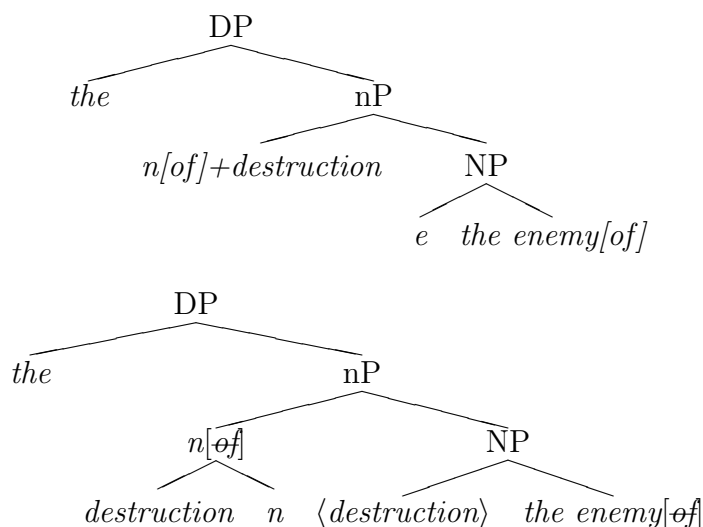
(141)



Two things are wrong with this structure: the $[of]$ feature on n has not been checked, and the Agent has not had its case feature valued.

Notice that we correctly rule in (140) with an interpretation for *the enemy* as Theme. In this case, we Merge *destruction* with the syntactic object *the enemy*. The $[of]$ feature on n values that on *the enemy*, and the structure is well formed:

(142)



We also capture the fact that a Saxon genitive may have an interpretation as a Theme, in a parallel fashion to the way we captured the fact that certain subjects could be themes: all of these constructions involve movement of the Theme into a higher position.

7.4.3 Possessors

In this section we briefly discuss the syntax of Possessors. We saw earlier that nouns which have no argument structure may still occur with possessors:

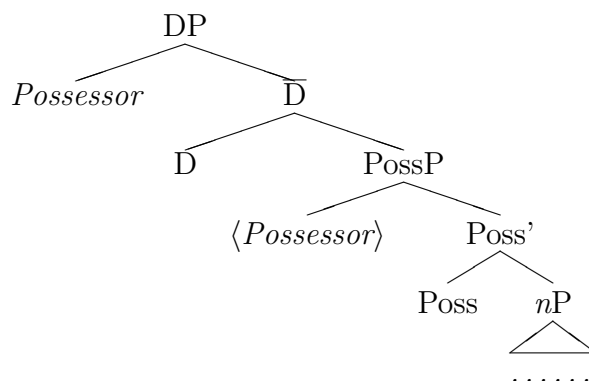
(143) Jenny's cat

(144) My penknife.

The surface syntax of possessors is just the same as other Saxon genitives: they are in the specifier of DP, and they have their case value assigned by the D head of DP. However, the Merge position of these categories is far from clear.

If the generalisation of the UTAH to the nominal domain is correct, possessors cannot be Merged as daughters of NP, or as specifiers of n , since these are positions which are reserved for particular θ -roles. An alternative proposal is to assume that there is an optional functional category, specified as part of the Hierarchy of Nominal Projections, and that Possessors are Merged in the specifier of this category. Such a structure would look as follows:

(145)



The function of this functional head is to introduce a semantic relationship between the element in its specifier, and the nP . Often this relationship is one of possession, but this is not necessary. An example like (146) can have a plethora of interpretations: it could be the book belonging to Anson, the one he's holding, the one that he drew the cover illustration for etc.

(146) Anson's book.

We will not motivate this approach to possessors any further. See the reference section for further details. We will also ignore the licensing of independent genitives. There is no clear consensus in the literature as to the proper analyses of these constructions.

Let us summarize our final position on the analysis of arguments within DP:

1. The UTAH generalises to the nominal domain. Agents within DP are generated in the specifier of a little n whose complement is NP. The head of NP raises into n , just as the head of VP raises into v . Little n optionally selects a specifier
2. The genitive marked DP which fills the specifier of genitive marked Ds (i.e. *s* and *every*) moves to that position from lower in the structure, driven by the strength of valued genitive case.
3. Possessor's are Merged in the specifier of a specialized optional functional head Poss, which establishes an underdetermined semantic relation. The Possessor checks its [case] feature with the [gen] feature of D, and moves to satisfy strength of this feature.

7.5 AP modification in DP

In previous sections, we have concentrated on the functional structure in DP and on the case checking and position of arguments. In this section we will look briefly at adjuncts in

DP. We will concentrate on the behaviour of adjectival phrases (APs) as exemplified in the following examples:

- (147) The **complicated** analysis
 (148) A **stunning** photograph of Jenny
 (149) Mary's **striking** resemblance to Sue

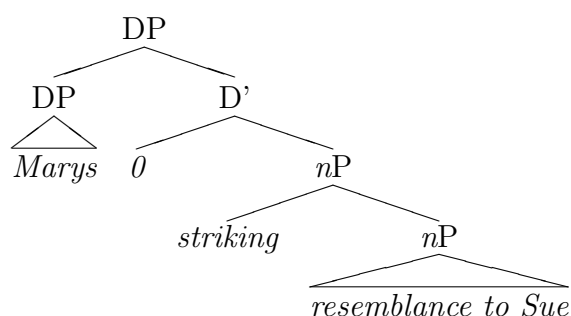
In both of these examples the determiner is separated from the head of the noun phrase by an adjective. The adjective itself can be modified by elements like *very*, *extremely*, *quite* etc., forming an adjectival phrase (AP). We will ignore the structure of AP itself, and concentrate on how APs are merged into DPs:

- (150) The [very complicated] analysis
 (151) An [extremely stunning] photograph of Jenny
 (152) Mary's [quite striking] resemblance to Sue

APs behave semantically as modifiers. They are not assigned theta-roles. Given this, there are two plausible approaches to take, within the framework of assumptions we have been developing: they are adjoined to some projection, or they are in the specifier of a functional (non- θ -related) head. We will pursue the former approach here, maintaining uniformity in our treatment of Adjunct in general. However, see the reference section for alternative approaches.

In this section, then, we will briefly explore the first option: that APs are adjoined to projections within the DP. Since the APs in the examples above occur between D and (the final position of) N, it is reasonable to assume that they are adjoined to *nP*, in the same way as adverbs are adjoined to *vP*:

(153)



This approach predicts that APs are optional, since the operation Adjoin is not an operation which is triggered by feature checking requirements, and furthermore, it predicts that APs may recurse, since the output of the adjunction operation is still an *nP*. Both of these predictions are confirmed:

- (154) Mary's resemblance to Sue
 (155) Ron's sleepy little rat

(156) Jenny's scraggy extremely demonic cat

APs clearly signal semantic modification of *nPs*, as do PPs:

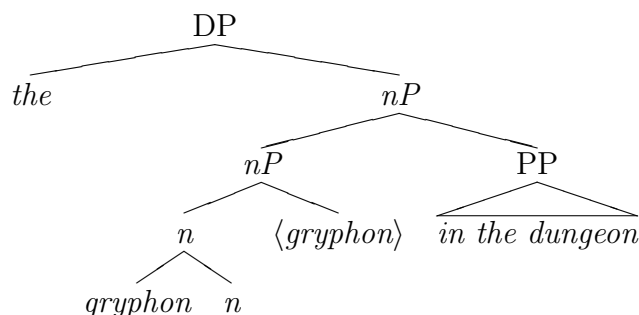
(157) The gryphon in the dungeon howled loudly.

(158) Ron's sleepy mouse in his pocket sneezed.

It is important to think carefully about the semantic relationship between a noun and a PP which occurs within the noun's DP. If the PP is an argument of the noun, then it must be in complement position. If the PP is not an argument, then it must be adjoined. In (157) and (158) the PP is clearly not an argument, since it simply signals the location of the entity picked out by the noun, so the PP is adjoined to the noun.

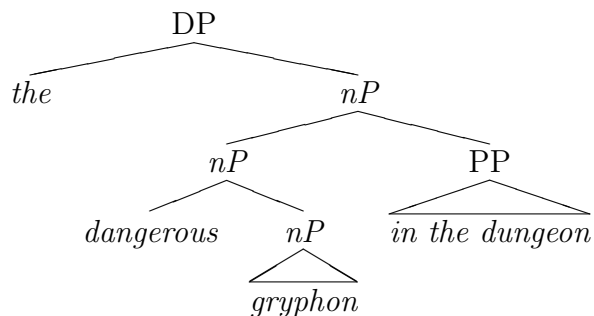
We represent the adjunction of PP as adjunction to *nP*, in much the same way as APs are adjoined to *nP*:

(159)

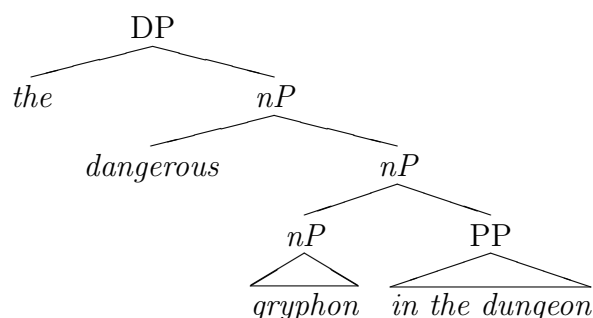


When we have both an AP and a PP adjoined to *nP*, there is usually no syntactic way of knowing which is joined lower. However, since adjunction corresponds semantically to modification, in (160) what is in the dungeon is a dangerous gryphon, while in (161), the thing that is dangerous is the gryphon in the dungeon:

(160)



(161)

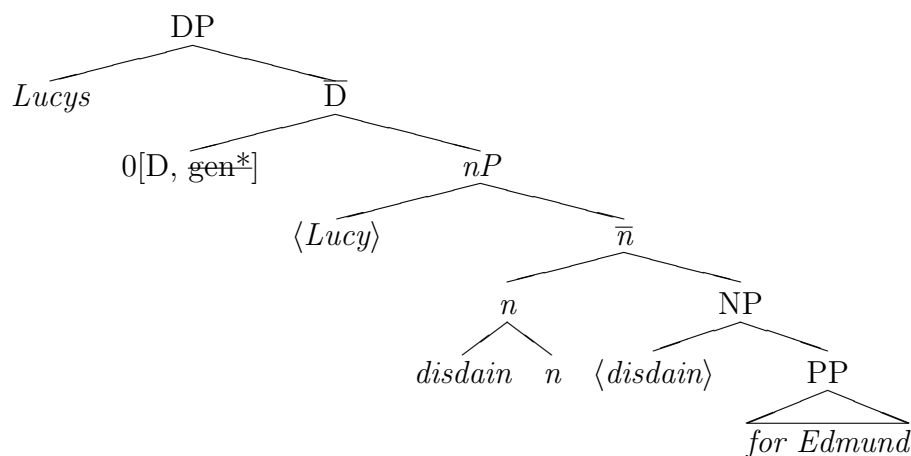


PPs like those just seen differ from cases like the following:

- (162) Mary's resemblance to Sue
- (163) Lucy's disdain for Edmund
- (164) Our discussion of porcupines

In each of these cases, the PP is an argument of the noun. These PPs are complements of N, Merged with N because of N's selectional P-feature:

(165)



One final thing to watch out for with PPs is that they can be complements and adjuncts to verbs as well as nouns. This means that, when a PP or a series of PPs comes at the end of the sentence, it is important to work out whether they belong to the object or to the verb. Take, for example, as case like the following:

- (166) Ron heard a discussion in the foyer

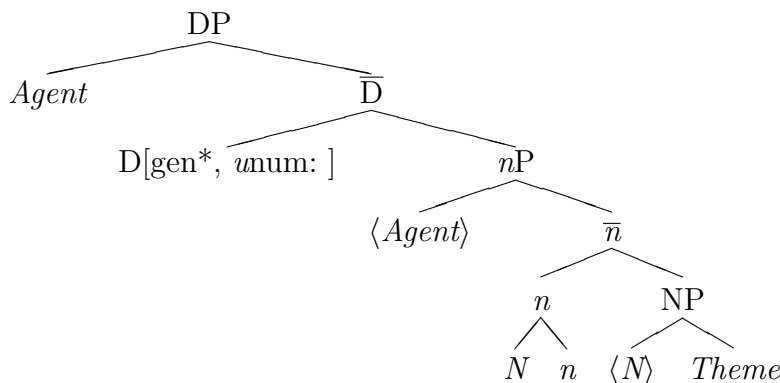
Here the PP *in the foyer* is not a semantic argument of *discussion*, and so it cannot be a syntactic complement - it must instead be adjoined somewhere. The question is whether it is adjoined to NP or to VP. In this case the sentence is ambiguous: the discussion itself might be located in the foyer, with Ron listening to it via a keyhole from a classroom, or the discussion might be taking place in a secret chamber, buried beneath the foyer. Ron's hearing of the discussion, however, occurred in the foyer. In the first case the PP is adjoined to NP, since it modifies the noun *discussion*, in the second case it is adjoined to VP, since it modifies the verb *heard*.

7.6 Order and Movement in DP

Before leaving DPs, we should note that the analysis we have developed leads to particular expectations about cross-linguistic variation in DPs. In chapters 5 and 6, we saw that some languages raised *v* to T (French), while others did not (English and Swedish), and that some languages allow the subject to remain in its base position (Scottish Gaelic), while others require it to raise (English and French). We encoded these differences via the varying strength properties of features on functional heads.

Our analysis of DP structure in English also depends upon functional heads and their featural properties. The structure of DP looks schematically as follows (abstracting away from possessors):

(167)



The [gen*] feature of D is strong, and forces movement of the closest DP whose case feature it values (in this case, the Agent, but it could instead have been a Possessor, or a Theme). The [unum:] feature on D encodes the number agreement relationship between D and the noun, and ensures that *each* combines with a singular noun, and *many* with a plural. We encoded this via Agree. This feature is weak in English.

Now, if we vary the strength of these features then we would expect to find languages where the *unum* is strong, and [gen] on D is weak. This would predict that the N+n complex would raise to D, and that the Agent would stay in situ.

In fact, such languages do exist, and the construction that displays these properties is known traditionally as the **construct state**. The construct state is found in Semitic languages, and I give examples here from Modern Hebrew:

(168) harisat ha-oyev 'et ha-'ir
 destruction the-enemy OM the-city
 'The enemy's destruction of the city.'

(169) tipul ha-Siltonot ba-ba'aya
 treatment the-authorities in-the-problem
 'The authorities' treatment of the problem.'

In each of these cases, the whole DP is interpreted as though it is definite, even though there seems to be no definite determiner as its head. What we find instead is that the noun head

appears before the Agent, and that the Agent appears before the object (OM in (168) stands for **object marker**, a special, preposition like element in Hebrew whose syntactic function seems to be to realise accusative case).

What is most interesting about the construct state is that no determiner is possible:

- (170) *ha-harisat ha-oyev 'et ha-'ir
 the-destruction the-enemy OM the-city
 'The enemy's destruction of the city.'

Furthermore, the order of the Agent and Theme is strict:

- (171) *harisat 'et ha-'ir ha-oyev
 destruction OM the-city the-enemy
 'The enemy's destruction of the city.'

These two properties are immediately accounted for on the analysis suggested above. If the $N+n$ complex raises to D, because [$unum$:] (or its value) is strong, then the lack of a D in the construct state can be explained by assuming that the $N+n$ complex 'fills' the D position in the structure (or alternatively, that the D bearing [$unum^*$] is null). If [gen] on D is weak, then we explain why the Agent appears after the noun rather than before. Finally, since the UTAH governs where Agents and Themes will be realised, the strict order of the arguments within the DP is predicted. Under this set of proposals, the construct state in Hebrew looks roughly as follows:

- (172)
-
- ```

graph TD
 DP --> D1["D[gen]"]
 DP --> nP["nP"]
 D1 --> n1["n[sing, acc]"]
 D1 --> D2["D[unum:sing*]"]
 n1 --> destruction["destruction"]
 n1 --> n["n"]
 D2 --> the_enemy1["the enemy"]
 nP --> D3["D[gen]"]
 nP --> nb["n̄"]
 D3 --> the_enemy2["the enemy"]
 nb --> destr_n["<destruction n>"]
 nb --> NP["NP"]
 NP --> destr["<destruction>"]
 NP --> DP_acc["DP[acc]"]
 DP_acc --> the_city["the city"]

```

The [ $acc$ ] feature on little  $n$  is analogous to the *of*-feature we proposed for English. The [ $gen$ ] feature of D checks and values the case feature of *the enemy*, but does not trigger it to raise, since [ $gen$ ] is weak. However, the [ $unum$ ] feature on D is strong, and triggers movement of little  $n$  (and everything inside it) to D.

This analysis barely scratches the surface of the complications we see in cross linguistic variation in DP structure, but it gives some indication of how the framework we are developing leads to expectations about the kinds of variation we might expect. See the reference section for further reading.



## 7.7 Summary

In this chapter we have developed the proposal that there is a significant parallelism between the clausal and nominal phrases. We have argued that a number of aspects of the syntax of nominal phrases receive explanations if we adopt the idea that these phrases are headed, not by a noun, but rather by an element of category D.

We proposed that D appeared even in nominal phrases which were apparently missing a determiner, and we developed a theory of null Ds, which captured the selectional properties inside DP, and allowed us to maintain our theory of c-selectional features in its most uncomplicated form.

We also argued that D could be a locus of case checking for arguments inside the DP, much as T is in TP. We proposed that D bears a strong [gen\*] feature in English, so that the specifier of certain Ds comes to be filled with a Saxon genitive.

We also argued that the UTAH extended to  $\theta$ -role assignment within DP, and proposed that, analogous to little  $v$ , DPs contained a little  $n$ , whose specifier receives an interpretation as an Agent. Little  $n$  is responsible for case checking of any Theme in the DP. The syntax we proposed allowed us to capture generalisations about the realisation of arguments within DP, specifically that Agents could not be realised as *of*-PPs, and that Themes and Agents could both be realised as Saxon genitives.

We tentatively suggested that true possessive DPs were generated in the specifier of an optional functional head, which we placed above little  $n$ P. Just as D[gen\*] enters into an Agree relation with the Agent, and triggers the Agent to move to the specifier of DP, it may instead enter into an Agree relation with a possessor, so that the possessor can end up having the same syntax as a raised Agent or Theme.

In order to explicitly state our analysis of DP structure, we extended the Hierarchy of Projections to cover the nominal domain:

- (173)    **Hierarchy of Projections:**  
           **Clausal:** T  $\rangle$  (Neg)  $\rangle$  (Perf)  $\rangle$  (Prog)  $\rangle$  (Pass)  $\rangle$   $v$   $\rangle$  V  
           **Nominal:** D  $\rangle$  (Poss)  $\rangle$   $n$   $\rangle$  N

We briefly looked at adjectival and prepositional modification in DPs, and proposed that these elements were adjoined to the little  $n$  projection. We noted that other analysis, which extend the Hierarchy of Projections yet further, are also possible for AP and PP modifiers.

Finally, we showed that our analysis of DPs extends in the expected way to constructions in other languages which look radically different. Once again, the differences reduce to parametric settings in feature strength of heads in the Hierarchy.

In this chapter more than in previous chapters we have left many loose ends: the interactions between possessors and Agents, the syntax of the independent genitive, alternative approaches to modification, and the potential of richer functional structure in the Hierarchy. This has been necessary so that this particular chapter does not grow out of all proportion. You will have a chance to explore some of these topics in the exercises, and I have included relevant references in the further reading section.

Once again, here are the most important phenomena we met, and the basics of their explanations:

| Phenomenon                                        | Explanation                                                                       |
|---------------------------------------------------|-----------------------------------------------------------------------------------|
| Complementary distribution of Ds                  | The Hierarchy of Projections extends to the nominal domain                        |
| Selectional relationships between Ds and Ns       | uninterpretable features on D, which are valued by features on N or <i>n</i>      |
| The appearance and position of arguments of nouns | Extension of the UTAH to the nominal domain, and case valuing by functional heads |
| The case of arguments inside DP                   | [gen*] on some Ds, and [of] on little <i>n</i>                                    |
| AP and PP modification                            | These elements are <i>n</i> P adjuncts                                            |
| Construct State phenomena                         | Parametric variation in the strength properties of [gen] and [ <i>unum</i> ]      |

## Exercises

### Exercise 1 — Gerunds

In English, there is a form of the verb called the **gerund**. This form is specified by suffixing *ing* to the verb. In this exercise, we will suppose that *ing* in English is ambiguous between being a little *n* bearing an [of] case feature, and a little *v* bearing an [acc] case feature.

#### Part A

Draw a tree, fully specified for case features, for each of following phrases in bold in the following sentences. Note that the phrases in bold have to be DPs, since they are in subject position.

- (1) **The reading of Shakespeare** satisfied me
- (2) **Reading Shakespeare** satisfied me

#### Part B

There now follow some datasets. Say for each data set whether it supports the tree you drew above, or whether it is problematic. Say why in each case:

DATASET 1

- (3) The constant reading of Shakespeare (satisfied me)
- (4) Constantly reading Shakespeare (satisfied me)
- (5) \*The constantly reading Shakespeare (satisfied me)
- (6) \*Constant reading Shakespeare (satisfied me)

## DATASET 2

- (7) The readings of Shakespeare (satisfied me)
- (8) \*Readings Shakespeare (satisfied me)

## DATASET 3

- (9) No reading of Shakespeare (satisfied me)
- (10) \*No reading Shakespeare (satisfied me)
- (11) \*Not reading of Shakespeare (satisfied me)
- (12) Not reading Shakespeare (satisfied me)

## DATASET 4

- (13) Every reading of Shakespeare (satisfied me)
- (14) \*Every reading Shakespeare (satisfied me)
- (15) Every reading of Shakespeare (satisfied me)
- (16) \*Every reading Shakespeare (satisfied me)

## DATASET 5

- (17) \*Having read of Shakespeare (satisfied me)
- (18) Having read Shakespeare (satisfied me)

Note that (17) here is grammatical on a different interpretation. It can mean having read about Shakespeare but it can't mean having actually read Shakespeare's works. The star in the example is indicating that this sentence is ungrammatical under the reading where Shakespeare is the Theme argument of *read*.

**Part C**

Gerunds can also have subjects:

## DATASET 6

- (19) Bill's reading of Shakespeare (satisfied me)
- (20) Bill's reading Shakespeare (satisfied me)

Say how you think these subjects are added onto the structures you drew in part A. Point out anything unexpected that is going on in the trees.

**Part D**

Now look at the following dataset. Here you can see that the little *v* version of *ing* can have a different kind of subject which is accusative. It is possible to capture this difference by positing a null determiner. Say what you think the selectional and case properties of this null determiner are.

## DATASET 7

- (21) \*Bill reading of Shakespeare (satisfied me)
- (22) Bill reading Shakespeare (satisfied me)

**Part E**

The null D approach to this data captures everything we have seen so far. The following dataset, however, shows that it is not the last word. Given the data in (23) and (24), explain why the last two examples are problematic.

- (23) \*Bill's reading Shakespeare and Maureen's singing Schubert satisfies me
- (24) Bill's reading Shakespeare and Maureen's singing Schubert satisfy me
- (25) Bill reading Shakespeare and Maureen singing Schubert satisfies me
- (26) \*Bill reading Shakespeare and Maureen singing Schubert satisfy me

**Exercise 2 — Trees for Ds**

Draw trees, fully annotated with case features, for the following DPs.

- (27) Her every wish
- (28) That book's cover
- (29) Each picture of Hermione's aunt's fish
- (30) The sultan's every desire's fulfilment
- (31) That girl's best friend's fluffy moustache

**Exercise 3 - Diagnostics for P**

A good test for telling whether a word is a preposition in English is that the word *right* may be inserted immediately before it:

- (32) I climbed up the tree.
- (33) I climbed right up the tree.

Apply this test in the following examples to the words in bold, which would all be traditionally analysed as prepositions. Can you see anything in common that holds the prepositions which fail this test? Think carefully about the  $\theta$ -roles assigned in the sentences:

- (34) I compared Ginger **to** Fred
- (35) We took the car **to** the town
- (36) I bought a book **about** Harry
- (37) We arrived **about** six o'clock
- (38) They depend **on** Mary.

- (39) They sat on Mary.
- (40) I thought **of** the moon
- (41) She took a picture **of** the hippogryph

Is there anything special about the preposition *of*?

### Exercise 4 — Adjunct or argument?

For the following examples, state whether the final PP is more likely to be construed with the verb or the object. If you think it is ambiguous, try to state as clearly as you can what the difference in meaning is:

- (42) They brought the hat to the teacher
- (43) No one could remove the blood from the wall
- (44) No one could remove the blood on the wall
- (45) The flying car hit the tree in the air
- (46) The elixir was hidden in a hole in the ground

### Exercise 5 — More tree drawing

Draw trees for the following DPs and sentences. Annotate the trees with bar-levels and phrasal status, but leave out selectional features, case features etc. You may assume that adjectives are always adjoined lower down than PP adjuncts.

- (47) Our discussion of previous students' failures in exams traumatised every nervous student in the class yesterday.
- (48) The continual implementation of modularization in this University demands constant attention.
- (49) Merlin's remarkable recovery from his hideous disease surprised the evil warlocks' chief curser.

Now give a derivation for one of these sentences.

## Further Reading

**Section 7.1:** An early discussion of how determiner like elements enter into the syntax of nominals can be found in Jackendoff (1977). See also chapter 4 of Haegeman and Guéron (1999) for a thorough textbook treatment. A good discussion of definiteness is Lyons (1997), and see Heim (1982) for a thorough theoretical treatment. For the semantics of quantifiers, see any good semantics textbook (Saeed 1997 provides a basic account, and de Swart 1998 and Heim and Kratzer 1998 give more theoretical approaches).

**Section 7.2:** A full length discussion of the question of whether determiners are heads can be found in Abney (1987). A more cross-linguistic perspective, focussing on the differences

between Germanic and Romance is Giorgi and Longobardi (1991). Szabolcsi (1983) provides a full discussion of the Hungarian data mentioned here, which is considerably more complex than what is sketched in the text (see also Szabolcsi 1994). There have been various theoretical extensions of the basic idea that nominal phrases are headed by D: for proper names see especially Longobardi (1994), and for pronouns see Cardinaletti and Starke (1999).

**Section 7.3:** An early discussion of how  $\theta$ -role assignment takes place in nominals can be found in Chomsky (1970). For later work see especially Grimshaw (1990). Barker (1991) discusses how possessors are integrated into the semantics of DPs.

**Section 7.4:** The analysis we adopted in the text is essentially an updating of Abney (1987) with some changes to fit the current state of the theory. A good collection of papers on the issues here is Alexiadou and Wilder (1998).

**Section 7.5:** The approach we adopted in the text, whereby APs adjoin to  $nP$ , is an update of a standard Government and Binding approach (see textbooks in GB theory, for example Haegeman 1991). Valois (1991) argues for this approach. The alternative view, that APs are specifiers of functional heads, is argued for by Cinque (1993), among others. See Svenonius (1993) and (Alexiadou 2001) for discussion.

**Section 7.6:** The construct state has received a lot of attention in the literature on Semitic syntax. See Borer (1999), Ouhalla (1991), Ritter (1991). Constructions very similar to the Semitic construct are also found in Celtic languages: see Duffield (1996) for Irish, and Rouveret (1991) for Welsh.